

Technical Datasheet — LNMO High-Voltage Smartphone Cell

Field	Value	Source
Rated capacity (Ah)	4.5 (nominal) / 6.372 (simulation-verified 1C)	parameter set + `r5_d11_1c.json`
Nominal voltage / voltage window (V)	4.17 / 2.5–4.7	`r5_d11_energy.json:midpoint_voltage_v` + parameter set
Rated energy (Wh)	26.48	`r5_d11_energy.json:energy_wh` (V·I time integral)
Energy density — gravimetric (Wh/kg)	596.5	`r5_d11_energy.json:energy_density_wh_kg` (electrolyte excluded)
Energy density — volumetric (Wh/L)	1260.2	`r5_d11_energy.json:energy_density_wh_l` (electrolyte excluded)
Maximum continuous discharge rate	1C (6.37 Ah)	`r5_d11_1c.json` (1C discharge protocol)
Fast-charge capability	4C CC charge, no plating (anode min +0.043 V), T_max 321.68 K	`r5_d11_4c.json` (4C_charge_45C + plating, lumped thermal)
Operating temperature range	25 °C–45 °C (simulated); upper bound = 4C-charge ambient	protocol conditions
Cycle life	**Not simulated beyond 100 cycles; SEI 193.6 nm after 100×1C cycles**	`r5_d11_aging.json`
Safety determination	no lithium plating at 4C; T_max 321.68 K ($\leq$ 323.15 K)	`r5_d11_4c.json`
DC resistance (Ω)	0.0105	`r5_d11_energy.json:dcr_ohm`
Dimensions	electrode 65 mm × 1580 mm (unwound); layer stack 204.6 μ m	parameter set + `r5_d11_energy.json`
Mass (g)	44.4 (electrolyte/casing excluded)	`r5_d11_energy.json:mass_kg`

Notes: gravimetric/volumetric energy density and mass use the contract-caliber formula (electrolyte and casing excluded, per `calc-energy`). Cycle life beyond 100 cycles is not simulated (no extended aging model run); the value shown is the 100-cycle SEI thickness.